

AWS CLOUD

(VIDEO STREAMING PLATFORM / OTT SERVICE)

AWS:

AWS means Amazon Web Services and it is a cloud computing platform.

It is used to store data and manage it using the internet.

We use AWS because it is easily accessible from anywhere, experiences no downtime even during massive viewer surges, and follows a pay-as-you-go model based on the bandwidth consumed.

AWS Cloud:

Cloud means storing video files and running media delivery networks on the internet instead of managing physical servers and hard drive racks in a local studio.

Amazon EC2:

Amazon EC2 stands for Elastic Compute Cloud.

It is like an online computer which provides massive processing power to run heavy transcoding software and video rendering applications that cannot be handled by standard laptops.

Use in Streaming App:

- I can use it to run the central user authentication and video catalog page.
- I can handle millions of concurrent user profile updates and search queries using EC2.
- I can use it to power the recommendation algorithm that suggests new shows to watch.

Amazon S3:

S3 stands for Simple Storage Service.

It is used to store files and large multimedia objects online securely.

Use in Streaming App:

- I can store these files:
 - Raw high-definition video files (4K/1080p source files)
 - Movie posters, thumbnails, and preview clips
 - Subtitle files in multiple languages (.srt / .vtt)
 - User profile avatars and cover images

Backup Files:

Use in Streaming App:

Instead of saving massive video directories on local office drives, I can store all movie files directly in S3. The application streams the video chunks to viewers directly from the cloud, ensuring we never run out of storage space when adding new seasons or movies.

AWS Lambda:

AWS Lambda runs code automatically when specific events happen.

There is no need to keep a server running continuously.

It is used to manage backend execution on-demand so we do not pay for idle computing power when nobody is interacting with the system.

Use in Streaming App:

I can use this in my project because users upload content or trigger actions randomly. Instead of running an expensive server 24/7, I can use AWS Lambda to run a script *only* when a new video is uploaded to convert it automatically into multiple resolutions (720p, 1080p) or to instantly generate a thumbnail image the exact second an upload completes.

AWS Security:

AWS provides robust security features to protect copyrighted media content and user privacy.

It provides:

- Secure User Sign-In & Verification
- DRM (Digital Rights Management) Integration
- Video Stream Link Expiration & Token Encryption
- Network Firewalls to Prevent Content Scraping
- Database Backups

Use in Streaming App:

- Only premium subscribers can access and request high-quality movie playbacks.
- The video streaming links expire automatically to prevent people from sharing or stealing the direct download links.
- Payment databases and sensitive user credentials remain fully protected.

AWS IAM (Identity and Access Management):

IAM is an AWS service used to control who (or which application) can access specific AWS cloud resources and what actions they are allowed to execute.

Use in Streaming App:

- **Platform Owner / Admin** → Full access to the entire AWS infrastructure, streaming distribution settings, and billing logs.
- **Content Creator / Studio Partner** → Can upload new video content files and manage movie descriptions, but cannot access user databases or change global configurations.
- **Viewer / End User** → Can only fetch video streams, create personal watchlists, and update their own profiles; they have zero direct access to the underlying storage buckets or servers.